

HARDWARE AND COMPONENTS BASICS

DAY 3: RAM AND STORAGE

1. Comparison of DDR3, DDR4 and DDR5 RAM

Feature	DDR3	DDR4	DDR5
Typical Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400 MT/s and higher
Voltage	1.5 V	1.2 V	1.1 V
Performance	Low compared with newer RAM	Better performance	Highest performance
Power Efficiency	Less efficient	More efficient than DDR3	Most power efficient
Typical Use	Older computers	Most common PCs and laptops	Newer and high-performance PCs
Main Advantage	Suitable for older systems	Good balance of speed and cost	Higher speed and bandwidth

Conclusion: DDR3 is an older type of RAM, while DDR4 provides better speed and power efficiency. DDR5 is the newest of the three and provides higher speed, bandwidth, and better performance for modern computers.

2. RAM Removal and Reinstallation

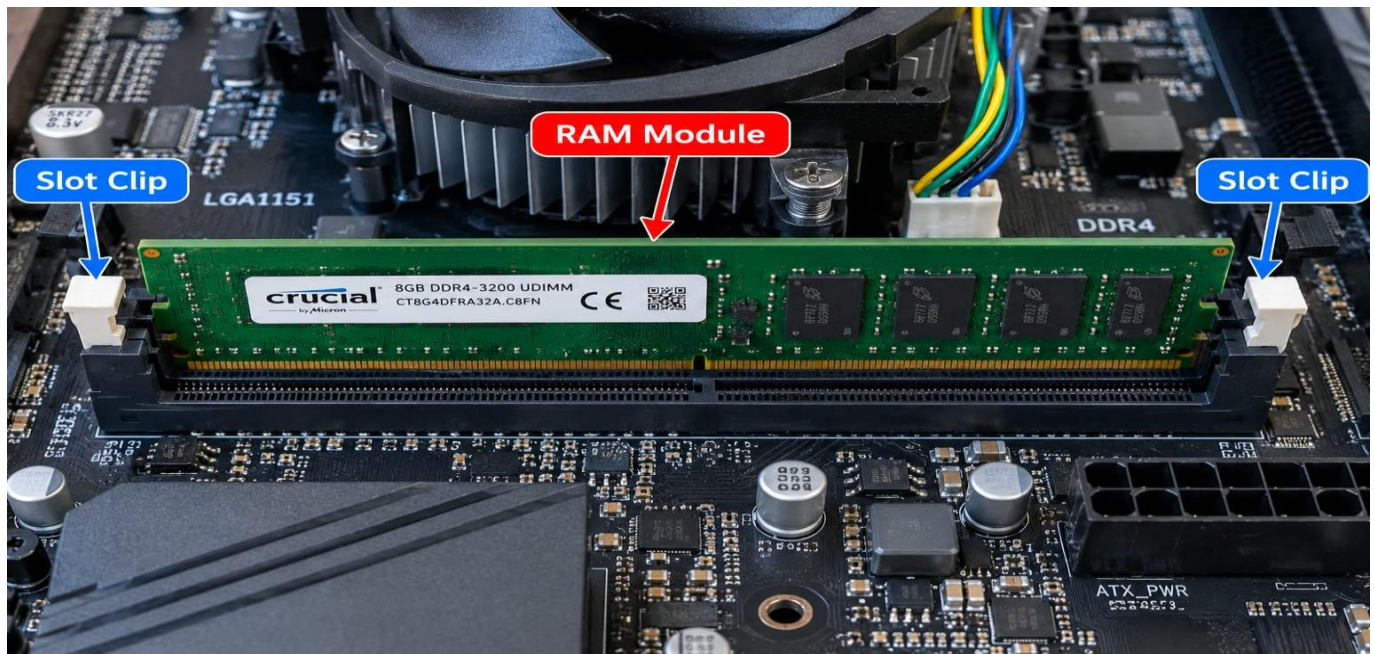
First, I disconnected the power from the computer/laptop and opened the back cover carefully. I located the RAM slot and checked how the RAM module was installed.

I gently opened the clips on both sides of the RAM slot and removed the RAM module. After checking the module and slot, I placed the RAM back into the same slot and pressed it down until the clips locked into place.

After reinstalling the RAM, I closed the back cover and turned on the system. The computer booted normally and the RAM was detected by the system.

Steps followed:

1. Disconnected the power from the device.
2. Opened the computer/laptop back cover.
3. Located the RAM slot.
4. Released the side clips and removed the RAM module.
5. Reinstalled the RAM module in the correct position.
6. Closed the back cover and switched on the system.
7. Checked that the system booted successfully and detected the RAM.



3. Three Key Differences Between HDD, SSD and NVMe

Storage Type	Difference 1	Difference 2	Difference 3	Suitable App / Use Case
HDD	Uses spinning magnetic disks	Slower read/write speed	Usually cheaper for large storage	Movies, photos and backups
SSD	Uses flash memory with no moving parts	Faster than HDD	More durable and quieter	Windows, Chrome and daily applications
NVMe	Uses PCIe interface	Much faster than SATA	Lower latency and higher performance	Gaming, video editing and large file transfers

HDD: HDD stands for Hard Disk Drive. It uses spinning magnetic disks to store data. It is slower than SSD and NVMe, but it is useful when a large amount of storage is required at a lower cost. Daily use: Storing movies, photos, documents and backups.

SSD: SSD stands for Solid State Drive. It uses flash memory and has no moving parts. It is much faster than an HDD and helps the operating system and applications start quickly. Daily use: Windows, Google Chrome and other regular applications.

NVMe: NVMe is a high-speed storage technology that normally uses the PCIe interface. It provides much higher read/write speeds and lower latency than traditional SATA storage. Daily use: Gaming, video editing and transferring large files.

4. Storage Drive Removal and Reinstallation

Before removing the storage drive, I disconnected the power from the device. I opened the back cover and located the storage drive.

In this example, the storage drive is an M.2 NVMe SSD. I carefully removed the retaining screw and took the drive out of the M.2 slot. I then placed the drive back into the same slot and secured it with the screw.

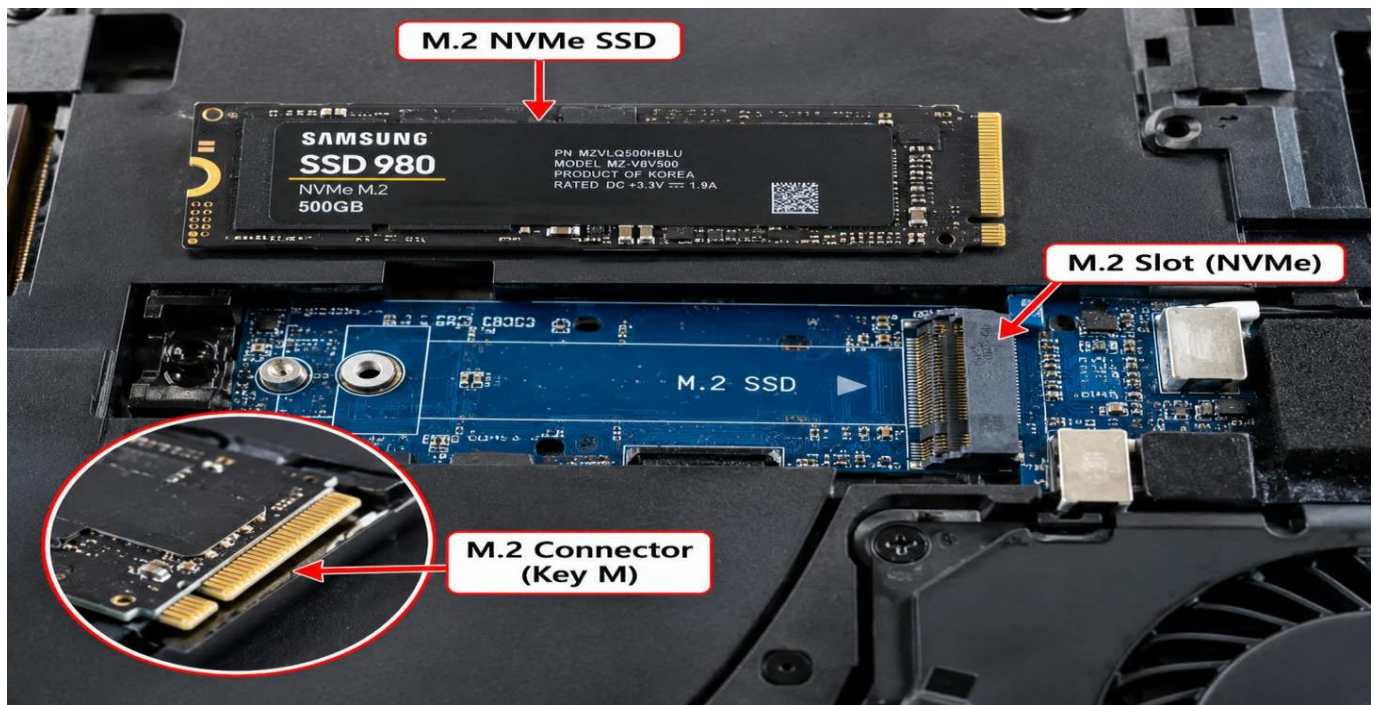
After closing the back cover, I powered on the device and checked the storage in the operating system. The storage drive was recognized successfully.

Steps followed:

1. Disconnected the power from the device.
2. Opened the back cover carefully.
3. Located the M.2 NVMe storage drive.
4. Removed the retaining screw.
5. Carefully removed the NVMe drive from the M.2 slot.
6. Reinstalled the drive in the same slot.
7. Secured it with the screw.
8. Closed the back cover and powered on the device.
9. Checked that the storage drive was recognized by the system.

Storage Connector Type: M.2 NVMe (PCIe)

Result: The system successfully recognized the storage drive after reinstallation and it was working normally.



Final Conclusion

In this practical, I learned about different types of RAM and storage devices. I compared DDR3, DDR4 and DDR5 RAM based on their speed and voltage. I also learned the differences between HDD, SSD and NVMe storage. Finally, I learned the basic process of removing and reinstalling RAM and an M.2 NVMe storage drive safely.